

Example SDI-12 commands

Name	Command	Response
Address Query	?!	a<CR><LF>
Change Address	aAb!	b<CR><LF>
Send Identification	aI!	allccccccmmmmmmvvvxxx...xx<CR><LF> Use the following information: a – sensor address (default =0) II – SDI-12 version compatibility (use = 14) ccccccc – 8 character, use 'ENG20009' mmmmm and vvv = 6 + 3 character, use your 9 digit student ID. Xxx – ignore this.
Start Measurement	aM!	atttn<CR><LF>
Send Data	aD0! . . . aD9!	a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF> a<values><CR><LF> or a<values><CRC><CR><LF>
Continuous Measurements	aR0! ... aR9!	a<values><CR><LF>

Examples			Notes
Name	Command	Response	
Address Query	?!	0	Address = 0
Change Address	0A1!	1	New address = 1
Start Measurement	0M!	00035	3 second with 5 parameters. BME680 - Temperature, Humidity, Pressure and Gas Sensor (4 parameters) BH1750FVI - Digital Light intensity Sensor (1 parameter)
Send Data	0D1! 0D2!	0+29.17+1004.46+57.64+66.06 0+270.00	BME680 parameters = 29.17°C (temperature), 1004.46 hPa (pressure), 27.64% (humidity), 66.06 KOHms (gas) 270.00 lx (light)
Send Identification	Send Identification	014ENG20009123456789	allccccccmmmmmmvvvxxx...xx<CR><LF> Use the following information: a – sensor address (default =0) II – SDI-12 version compatibility (use = 14) ccccccc – 8 character, use 'ENG20009' mmmmm and vvv = 6 + 3 character, use your 9 digit student ID. Xxx – ignore this.
Continuous Measurements	0R0!	0+29.17+1004.46+57.64+66.06+270.00	BME680 - Temperature, Humidity, Pressure and Gas Sensor (4 parameters) BH1750FVI - Digital Light intensity Sensor (1 parameter)